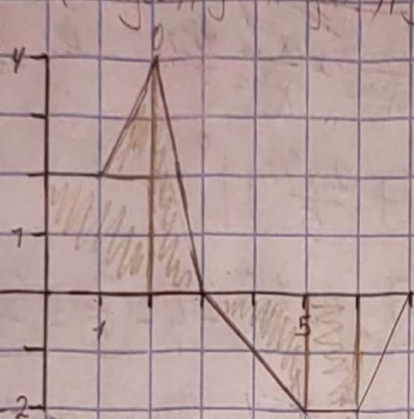


Tarea Sección 5.3

3) Sea $g(x) = \int_0^x f(t) dt$

a) Evalúe $g(0)$, $g(1)$, $g(2)$, $g(3)$ y $g(6)$



$$g(0) = \frac{d}{dx} \int_0^x f(t) dt = f(0) = 0$$

$$g(1) = \frac{d}{dx} \int_0^x f(t) dt = f(1) = 2$$

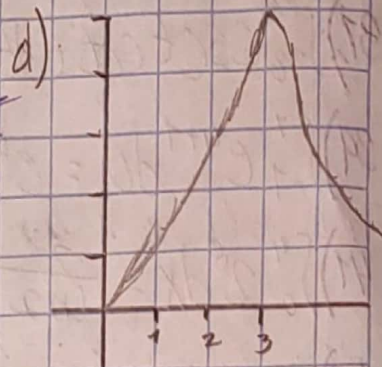
$$g(2) = \frac{d}{dx} \int_0^x f(t) dt = f(2) = 5$$

$$g(3) = \frac{d}{dx} \int_0^x f(t) dt = f(3) = 7$$

$$g(6) = \frac{d}{dx} \int_0^x f(t) dt = f(6) = 3$$

b) ¿En qué intervalo g es creciente? $(0, 3)$

c) ¿Dónde tiene g un valor mínimo? $x = 3$



9) $g(x) = \int_1^x \frac{1}{t^3+1} dt$

$$\frac{d}{dx} \int_1^x \frac{1}{t^3+1} dt = \frac{1}{x^3+1}$$

13) $h(x) = \int_1^{e^x} \ln t dt$

$$\frac{d}{dx} \int_1^{e^x} \ln t dt = \left[\frac{d}{du} \int_1^u \ln t dt \right] \frac{du}{dx} = e^x \frac{d}{du} \int_1^u \ln t dt$$

$$u = e^x$$

$$du = e^x dx$$

$$\frac{du}{e^x} = dx$$

$$e^x (\ln u = e^x \ln(e^x) = x e^x)$$

15) $y = \int_1^{3x+2} \frac{t}{1+t^3} dt = \frac{d}{dx} \int_1^{3x+2} \frac{t}{1+t^3} dt$

$$u = 3x+2$$

$$du = 3 dx$$

$$\frac{du}{3} = dx$$

$$\frac{d}{dx} \int_1^{3x+2} \frac{t}{1+t^3} dt = 3 \frac{d}{du} \int_1^u \frac{t}{1+t^3} dt = \frac{3u}{1+u^3} =$$

$$\frac{3(3x+2)}{1+(3x+2)^3}$$

$$79) \int_1^3 (x^2 + 2x - 4) dx = \left(\frac{x^3}{3} + \frac{2x^2}{2} - 4x \right) \Big|_1^3 = \left[\frac{3^3}{3} + \frac{2(3)^2}{2} - 4(3) \right] - \left[\frac{1^3}{3} + \frac{2(1)^2}{2} - 4(1) \right]$$

$$(9 + 9 - 12) - \left(\frac{1}{3} + 1 - 4 \right) = 6 - \left(-\frac{8}{3} \right) = \frac{26}{3}$$

$$29) \int_1^4 \frac{2+x^2}{\sqrt{x}} dx = \int_1^4 \frac{2}{\sqrt{x}} + \int_1^4 \frac{x^2}{\sqrt{x}} = \int_1^4 \frac{2}{x^{1/2}} + \frac{x^2}{x^{1/2}}$$

$$\int_1^4 (2x^{-1/2} + x^{3/2}) dx = \frac{2x^{1/2}}{\frac{1}{2}} + \frac{x^{5/2}}{\frac{5}{2}} = \left(4x^{1/2} + \frac{2x^{5/2}}{5} \right) \Big|_1^4 =$$

$$\left[4(4)^{1/2} + \frac{2(4)^{5/2}}{5} \right] - \left[4(1)^{1/2} + \frac{2(1)^{5/2}}{5} \right] = 8 + \frac{64}{5} - 4 - \frac{2}{5} = \frac{82}{5}$$

$$37) \int_{\pi/6}^{\pi/2} \csc t \cot t dt = -\csc t \Big|_{\pi/6}^{\pi/2} = -(\csc \frac{\pi}{2} - \csc \frac{\pi}{6}) = 1$$

$$39) \int_{-1}^1 e^{u+1} du = e^{u+1} \Big|_{-1}^1 = e^{1+1} - e^{-1+1} = e^2 - e^0 = e^2 - 1$$

$$41) \int_0^4 2^x dx = \frac{2^x}{\ln 2} \Big|_0^4 = \frac{(2^4 - 2^0)}{\ln(2)} = \frac{15}{\ln(2)}$$

$$47) y = 4 - x^2$$

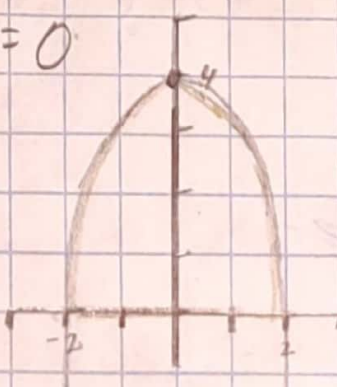
$$y = 0$$

$$-x^2 + 4 = 0$$

$$-x^2 = -4$$

$$x = \pm \sqrt{4}$$

$$x = \pm 2$$



$$\int_{-2}^2 -x^2 + 4 = \left(-\frac{x^3}{3} + 4x \right) \Big|_{-2}^2$$

$$\left(-\frac{(2)^3}{3} + 4(2) \right) - \left(-\frac{(-2)^3}{3} + 4(-2) \right) =$$

$$\frac{16}{3} - \left(-\frac{16}{3} \right) = \frac{32}{3}$$

$$59) g(x) = \int_{2x}^{3x} \frac{u^2-1}{u^2+1} du = \int_{2x}^0 \frac{u^2-1}{u^2+1} + \int_0^{3x} \frac{u^2-1}{u^2+1} = -\int_0^{2x} \frac{u^2-1}{u^2+1} + \int_0^{3x} \frac{u^2-1}{u^2+1}$$

$$\frac{d}{dx} \int_0^{3x} \frac{u^2-1}{u^2+1} - \frac{d}{dx} \int_0^{2x} \frac{u^2-1}{u^2+1} = \left(\frac{d}{dm} \right) \int_0^m \frac{u^2-1}{u^2+1} - \left(\frac{d}{dn} \right) \int_0^n \frac{u^2-1}{u^2+1} =$$

$$3 \frac{d}{dm} \int_0^m \frac{u^2-1}{u^2+1} du - 2 \frac{d}{dn} \int_0^n \frac{u^2-1}{u^2+1} du = \frac{3m^2-1}{m^2+1} - \frac{2n^2-1}{n^2+1} =$$

$$\frac{3(3x)^2-1}{(3x)^2+1} - \frac{2(2x)^2-1}{(2x)^2+1} = \frac{3(9x-1)}{9x^2+1} - \frac{2(4x-1)}{4x^2+1}$$

$$m = 3x$$

$$dm = 3dx$$

$$\frac{dm}{3} = dx$$

$$n = 2x$$

$$dn = 2dx$$

$$\frac{dn}{2} = dx$$

$$(01) F(x) = \int_x^{x^2} e^{t^2} dt = \frac{d}{dx} \int_x^{x^2} e^{t^2} dt = \frac{d}{dx} \int_x^0 e^{t^2} dt + \frac{d}{dx} \int_0^{x^2} e^{t^2} dt =$$

$$- \frac{d}{dx} \int_0^x e^{t^2} dt + \frac{d}{dx} \int_0^{x^2} e^{t^2} dt = \left[\frac{d}{dx} \int_0^u e^{t^2} dt - \frac{d}{dx} \int_0^x e^{t^2} dt \right] = \frac{d}{dx} \int_0^{x^2} e^{t^2} dt = 2x \frac{d}{dx} \int_0^{x^2} e^{t^2} dt$$

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$\frac{du}{2x} = dx$$

$$2x e^{u^2} - e^{u^2} = 2e^{x^4} - e^{x^2}$$

$$e^{u^2} = e^{(x^2)^2} = e^{x^4}$$

$$(03) y = \int_{\cos x}^{\sin x} \ln(1+2v) dv = \frac{d}{dx} \int_{\cos x}^{\sin x} \ln(1+2v) dv = \frac{d}{dx} \int_{\cos x}^0 \ln(1+2v) dv + \frac{d}{dx} \int_0^{\sin x} \ln(1+2v) dv =$$

$$- \frac{d}{dx} \int_0^{\cos x} \ln(1+2v) dv + \frac{d}{dx} \int_0^{\sin x} \ln(1+2v) dv = \frac{d}{dx} \int_0^{\sin x} \ln(1+2v) dv - \frac{d}{dx} \int_0^{\cos x} \ln(1+2v) dv$$

$$\left[\frac{d}{dn} \int_0^n \ln(1+2v) dv - \frac{d}{dm} \int_1^m \ln(1+2v) dv \right] =$$

$$- \cos x \frac{d}{dn} \int_0^n \ln(1+2v) dv + \sin x \frac{d}{dm} \int_1^m \ln(1+2v) dv =$$

$$\cos x (\ln(1+2n) + \sin x \ln(1+2m)) dv =$$

$$\cos x (\ln(1+2\sin x) + \sin x \ln(1+2\cos x))$$

$$\begin{cases} m = \cos x \\ dm = -\sin x dx \\ \frac{dm}{-\sin x} = dx \end{cases}$$

$$\begin{cases} n = \sin x \\ dn = \cos x dx \\ \frac{dn}{\cos x} = dx \end{cases}$$